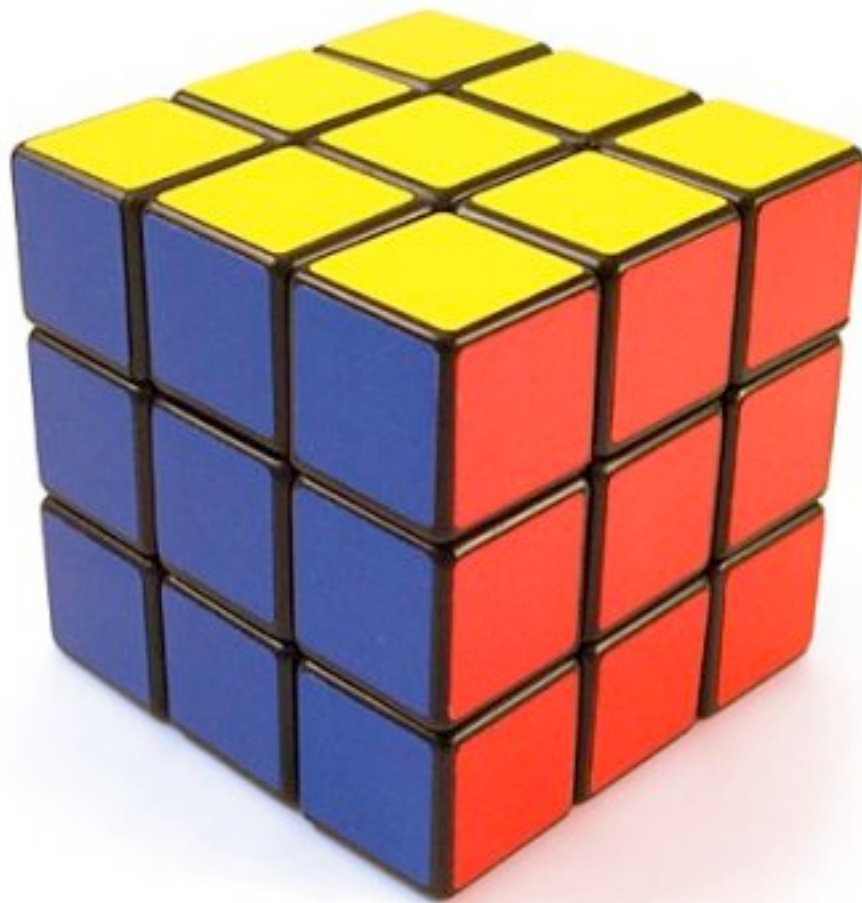


How to solve the Rubik's Cube



Solving Outline

- First Layer
 - White Cross
 - White Corners
- Second layer
 - Edges
- Last Layer
 - Orient Edges (Yellow Cross)
 - Orient Corners
 - Permute Corners
 - Permute Edges

General Thoughts and Tips

- First of all, this will be much easier if you get yourself a good speedcube. They move much smoother than Rubik's brand cubes, and my method relies heavily on muscle memory, so the smoother the cube the better. (google "speed cubes." There are lots of options out there and they're not too expensive).
- The method I've come up with requires you to only memorize 6 algorithms. Try not to memorize them with your mind, but with your hands. This is what I mean by "muscle memory." Do not move on to the next step until you can perform the current step without even thinking about it. The algorithms have a symmetry and a poetry to them that your hands will absolutely memorize with enough repetition.

Lesson 1

Notation

R



R'



U



U'



F



F'



Lesson 2

Basic Algorithms

Basic Algorithms

(R U R' U')

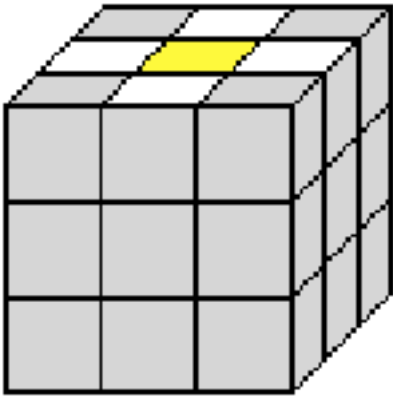
(R' F R F')

- Get very fast with these algorithms. Seriously, don't even move on until you can do these extremely fast without even thinking about it.
- If you do either of them six times in succession, the cube will return to its original state.
- These two algorithms (with very slight variations) will be all you need to memorize until step 7.

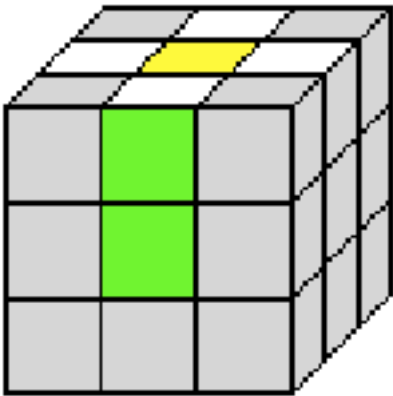
Lesson 3

First Layer Part 1: White Cross

Get all of the white edge pieces next to the yellow center piece like this:



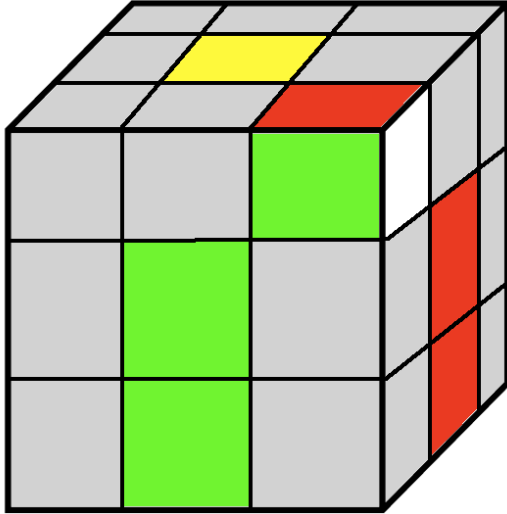
Now line up the other color on one of the white edge pieces with the center piece that has the same color. Like this:



Turn the F face 180° (In cubing notation this would be F2). Repeat the same process for the other three white edge pieces.

Lesson 4

First Layer Part 2: White Corners



Find a corner piece with white on it in the U layer. In this case it's the Green-Red-White corner. Put the piece above where it needs to go and on the right side of the cube.

Perform the following algorithm until the piece is in the correct spot **and** correctly oriented:

(R U R' U')

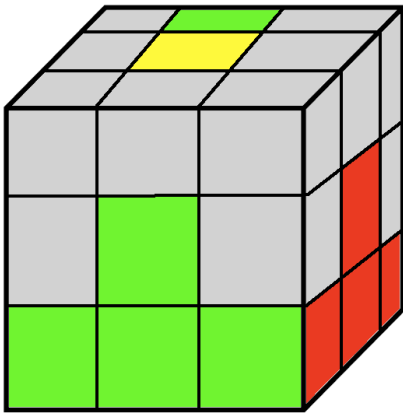
You will have to do it one, three, or five times for each white corner piece.

The algorithm shouldn't look like 4 moves to you at this point. It should look and feel like a single move that you can perform from muscle memory.

Lesson 5

Second Layer (Edges)

The white face should now be completely solved. Now it's time to insert the second layer edges.

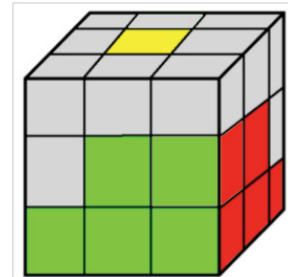


Find an edge piece in the top layer that does NOT have yellow on it. In this case it is the Green-Red edge. Put the piece in the back of the cube, and the place where it needs to go on the front-right of the cube, just like you see above.

Perform the following algorithm:

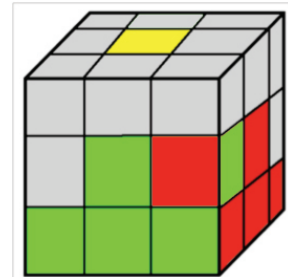
(R' F R F') (R U R' U')

If the edge was inserted correctly oriented, like so, you may move on.



If, however the edge is inserted with the wrong orientation, it will look like it does here.

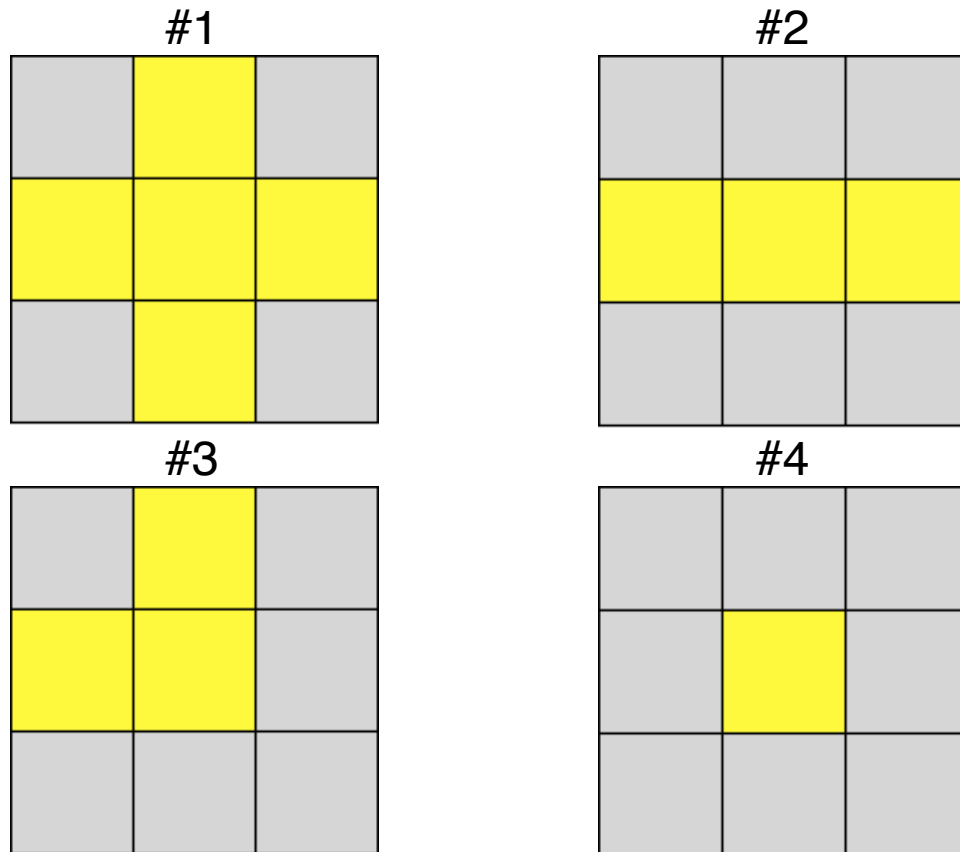
Simply perform the same algorithm as above, then twist the top face 180° (U2), then perform the algorithm again.



Lesson 6

Last Layer Part 1: Yellow Cross

Ignoring the corner pieces, the yellow edges will be oriented in one of the following patterns:



If you have case #1, you already have the yellow cross, so you can skip this step. In all other cases, perform the following algorithm and then reassess your situation.

F (R U R' U') F'

If you had case #4, you will have to now twist the top layer so that the “L” is oriented as it appears in case #3. If you had case #3, you will need to now twist the top layer so that the “line” is oriented as it appears in case #2. When you get the yellow cross you can stop and move on to the next step.

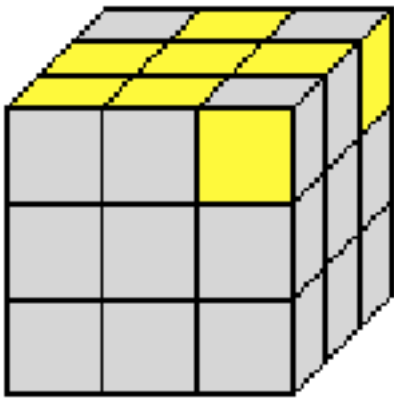
Lesson 7

Last Layer Part 2: Orient Corners

Once you have the yellow cross, you basically want to perform the following algorithm until you see a particular pattern. This is the algorithm:

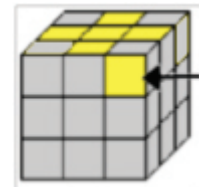
(R U R' U) (R U² R')

And the pattern you're looking for looks like this:



Once you see this pattern perform the algorithm one more time and all of the yellow stickers should be facing up.

There are many ways the yellow stickers can be arranged once you have the yellow cross, so it would get too confusing to get into specific cases. Just play around with the algorithm above until you see the “fish” pattern. As a general rule, you should favor having a yellow sticker facing you on the right hand before performing the algorithm.

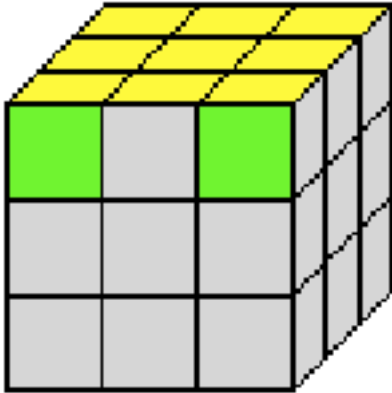


You will, however, run into cases where you find yourself in an “infinite loop.” You really just have to use trial and error to find the “fish” pattern, but you’ll never need to use any other algorithm to orient the last layer corners.

Lesson 8

Last Layer Part 3: Permute Corners

What you're looking for now I like to call "headlights."
Two corner stickers on the same face of the cube,
both in the last layer that are the same color. Like this:



- If you don't see headlights anywhere, do the algorithm below and headlights will appear.
- If you see headlights on every side, the corners are already permuted, so you can skip this step.
- Sometimes the headlights will have the same color in between them, sometimes they will not. It doesn't matter.

Put the headlights so they're facing you and perform the following algorithm:

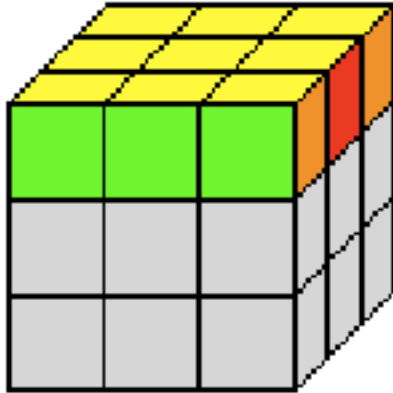
(R' U2) (R U2 R') F (R U R' U') (R' F' R2)

Note: "R2" means, turn the right face 180°, like you did to the front face in lesson 3.

Lesson 9

Last Layer Part 4: Permute Edges

You should now have headlights on all four sides. Look for the side that has three blocks of the same color like this:



If there is no side with three blocks of the same color, that's ok. Just use the same algorithm I'm about to tell you and it will create a side with three blocks of the same color.

Put the side with three blocks of the same color facing you (like it looks in the above image), and perform the following algorithm:

$(R' U R') (U' R' U') (R' U) (R U R^2)$

You will have to do it once or twice and the cube will be solved!